

“PAPER_{0:D3}” -f [int]# PAPER #82 □ Black Hole Evaporation Timescales: UQFF Corrections

Title: UQFF-Corrected Black Hole Evaporation Timescales: Stellar Mass Through Primordial Black Holes

Author: Daniel T. Murphy

Framework: UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57)

Date: March 7, 2026

Source Data: validate_hawking_temperature.py (Tests 2, 6), CondensedPhysics.py simulate_evaporation()

Index Slot: §1.11 Black Hole Physics & Hawking Radiation,
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Index Slot: §1.11 Black Hole Physics & Hawking Radiation, PAPER_082

Abstract

Black hole evaporation timescales are computed via the Stefan-Boltzmann law applied to UQFF-modified Hawking radiation. The standard result $t_{\text{evap}} = 5120\pi G^2 M^3 / (\hbar c^4)$ is modified by the $T_{\text{UQFF}}/T_{\text{H}} = 0.99$ ratio. This changes evaporation rates by a factor of $(T_{\text{UQFF}}/T_{\text{H}})^4 = 0.96$, extending evaporation timescales by $\sim 4\%$ for all black holes. The validate_hawking_temperature.py evaporation simulation (Test 6) confirms mass evolution for primordial BHs at 10^{10} kg over 100 timesteps.

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1. Evaporation Timescale Formula

Standard GR

$$t_{\text{evap}}^{\text{GR}} = \frac{5120\pi G^2 M_0^3}{\hbar c^4}$$

UQFF-Corrected

$$t_{\text{evap}}^{\text{UQFF}} = \frac{t_{\text{evap}}^{\text{GR}}}{(T_{\text{UQFF}}/T_H)^4} = \frac{t_{\text{evap}}^{\text{GR}}}{0.99^4} = t_{\text{evap}}^{\text{GR}} \times 1.041$$

UQFF evaporation timescale is 4.1% longer than GR □ black holes are slightly more stable in the UQFF vacuum.

2. Evaporation Timescales: Full Table

System	M0	t_evap_GR	t_evap_UQFF	Survives Universe
Sgr A*	4×10 ⁶ M?	8.7×10 ⁸ s	9.1×10 ⁸ s	? Yes
M87*	6.5×10 ⁷ M?	3.8×10 ²⁵ s	4.0×10 ²⁵ s	? Yes
Stellar BH	10 M?	2.1×10 ⁷⁴ s	2.2×10 ⁷⁴ s	? Yes
Primordial BH	5.7×10 ¹¹ kg	4.35×10 ¹⁷ s = t_U	4.52×10 ¹⁷ s	Borderline
Primordial BH	1×10 ¹⁰ kg	2.3×10 ¹² s (73 kyr)	2.4×10 ¹² s	? Evaporated

The validate_hawking_temperature.py Test 2 confirms: - Stellar BH (10 M?): survives_universe = True ? - Test 6 simulation: M_initial = 10¹⁰ kg, 100 steps, mass_lost_fraction computed ?

3. Mass Evolution Simulation

From simulate_evaporation(M_initial = 10¹⁰ kg, dt = 10¹⁰ s, n_steps = 100):

$$\frac{dM}{dt} = -\frac{k_{\text{UQFF}}}{M^2}, \quad k_{\text{UQFF}} = \frac{\hbar c^4 (T_{\text{UQFF}}/T_H)^4}{15360\pi G^2}$$

With T_UQFF/T_H = 0.99: k_UQFF = 0.96 × k_GR

At t = 100 × 10¹⁰ s = 10¹² s: - M_final □ M_initial × (1 - t/t_evap)^{1/3} = 10¹⁰ × (1 - 10¹²/2.4×10¹²)^{1/3} - M_final □ 10¹⁰ × 0.583^{1/3} □ 8.35×10⁹ kg - Mass lost fraction □ **16.5%** over first 10¹² s

Arrays times[], masses[], temperatures_H[] all have matching lengths (validate Test 6). ?

4. UQFF Mass Evolution Equation

The UQFF modifies the mass loss rate through the vacuum buoyancy correction. During late-stage evaporation ($M \gtrsim M_{\text{Planck}}$):

$$\frac{dM_{\text{UQFF}}}{dt} = -\frac{k_{\text{UQFF}}}{M^2} + \frac{g_{\text{Buoyant}} \times V_{\text{BH}}}{c^2}$$

The buoyancy term: $g_{\text{Buoyant}} \times V_{\text{BH}} / c^2 = \rho_{\text{vac}} \times 1055 \times (4/3)\pi r_{\text{S}}^3 / c^2 \sim 10^{-8} \text{ kg/s}$ negligible vs the thermal term at all masses above Planck mass.

Summary

Parameter	GR Value	UQFF Value	Change
Evaporation factor	k_{GR}	$0.96 \times k_{\text{GR}}$	-4%
Timescale t_{evap}	t_{GR}	$1.041 \times t_{\text{GR}}$	+4.1%
Stellar BH survival	Yes	Yes	Unchanged
Primordial threshold mass	$5.7 \times 10^{11} \text{ kg}$	$5.5 \times 10^{11} \text{ kg}$	-3.5%
Test 2	survives = True	Confirmed	? PASS

Source: `validate_hawking_temperature.py` Tests 2+6, `simulate_evaporation()` | $\rho = 0.0005/\text{day}$ | $[\text{SSq}] = 0.57$.Groups[1].Value "PAPER_{0:D3}" -f \$n

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Black hole evaporation timescales are computed via the Stefan-Boltzmann law applied to UQFF-modified Hawking radiation. The standard result $t_{\text{evap}} = 5120\pi G^2 M^3 / (\hbar c^4)$ is modified by the $T_{\text{UQFF}}/T_{\text{H}} = 0.99$ ratio. This changes evaporation rates by a factor of $(T_{\text{UQFF}}/T_{\text{H}})^4 = 0.96$, extending evaporation timescales by $\sim 4\%$ for all black holes. The `validate_hawking_temperature.py` evaporation simulation (Test 6) confirms mass evolution for primordial BHs at 10^{10} kg over 100 timesteps.

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Timescale t_{evap}	t_{GR}	$1.041 \times t_{\text{GR}}$	+4.1%
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UQFF computed: Eddington luminosity UQFF correction = 1 - [SSq]×exp(-?×?t) = 1 - 5.7e-1 × exp(-2.9e-4) = 4.3e-1; F_U at event horizon = 2.0e+18 m/s².

Appendix: UQFF Production Framework Reference (v4.75+)

Added by upgrade_early_whitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
$\hat{\Gamma}^0$	$5.0 \times 10^{-10} \text{ day}^{-1}$	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
$\hat{\Gamma}^2_i$	0.60 ± 0.61	Buoyancy coupling coefficient
k_{DPM}	1.5	Ug1 DPM-dipole coupling
k_{out}	1.2	Ug2 outer-bubble charge coupling
k_{str}	1.8	Ug3 string-rotation coupling
k_{vac}	2.0	Ug4 vacuum-concentration coupling
$\hat{I} \cdot$	10 cm^2	Inertia tensor scale
$E_{\text{react}}(0)$	10 MeV	Reference reactive energy

A.2 F_U Master Equation (Complete 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 \text{igl}[\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}} \text{igr}]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	compute_Ug1_SOURCE4 / compute_Ug1()
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
$\hat{\Gamma}^0 \hat{\Gamma}^2_i \hat{\Gamma}^0 \hat{\Gamma}^2_i$	4th-order dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER_420):
 $\hat{I}_{\text{c}}=10\hat{\mu}\hat{\text{A}}^1\hat{\mu}^\circ$, $\hat{I}_{\text{q}}=10\hat{\mu}\hat{\text{A}}^1\hat{\text{A}}^2$, $\hat{I}_{\text{a}}=10\hat{\mu}\hat{\text{A}}^1\hat{\text{A}}^1$, $\hat{I}_{\text{f}}=10\hat{\mu}\hat{\text{A}}^1\hat{\text{A}}^3$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \text{imesigl}(1 + 10^{13} \Theta(ho_{SCm} - ho_c)igr) \text{imesigl}(1 + A_q \cos(\Delta\omega t)igr)$$

Symbol	Value	Description
$\hat{I}_{\text{c}}$	$10\hat{\text{A}}^1\hat{\mu}\text{ kg/m}\hat{\text{A}}^3$	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\hat{I}_{\text{f}}$	$2\hat{\mu}/(434\hat{\text{A}}\cdot365.25)\text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_{sum} + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, $\hat{\mu}!$)	Multi-scale field interactions
Buoyant	$\hat{I}_{\text{c}}^2 \hat{\mu} \hat{\text{A}} \text{ Ubi}$	Expanding nebulae, stellar winds
Superconductive	$\hat{\mu} \hat{\text{A}} (1 + 10\hat{\text{A}}^1\hat{\text{A}}^3\hat{\text{A}}\cdot f_{\text{H}})$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, CondensedPhysics.py, and CondensedPhysics2.py.